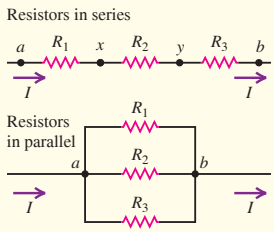
CHAPTER 26 SUMMARY

Resistors in series and parallel: When several resistors $R_1, R_2, R_3, \ldots$ are connected in series, the equivalent resistance R_{eq} is the sum of the individual resistances. The same *current* flows through all the resistors in a series connection. When several resistors are connected in parallel, the reciprocal of equivalent resistance R_{eq} is the sum of the reciprocals of the individual resistances. All resistors in a parallel connection have the same *potential difference* between their terminals. (See Examples 26.1 and 26.2.)

$$R_{eq} = R_1 + R_2 + R_3 + \cdots \quad \text{(resistors in series)} \tag{26.1}$$

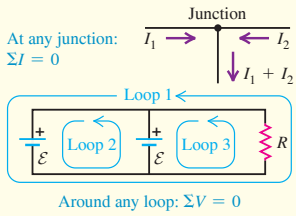
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots \quad \text{(resistors in parallel)} \tag{26.2}$$



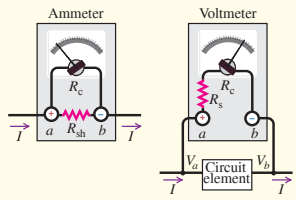
Kirchhoff's rules: Kirchhoff's junction rule is based on conservation of charge. It states that the algebraic sum of the currents into any junction must be zero. Kirchhoff's loop rule is based on conservation of energy and the conservative nature of electrostatic fields. It states that the algebraic sum of potential differences around any loop must be zero. Careful use of consistent sign rules is essential in applying Kirchhoff's rules. (See Examples 26.3–26.7.)

$$\sum I = 0 \quad \text{(junction rule)} \tag{26.5}$$

$$\sum V = 0 \quad \text{(loop rule)} \tag{26.6}$$



Electrical measuring instruments: In a d'Arsonval galvanometer, the deflection is proportional to the current in the coil. For a larger current range, a shunt resistor is added, so some of the current bypasses the meter coil. Such an instrument is called an ammeter. If the coil and any additional series resistance included obey Ohm's law, the meter can also be calibrated to read potential difference or voltage. The instrument is then called a voltmeter. A good ammeter has very low resistance; a good voltmeter has very high resistance. (See Examples 26.8–26.11.)

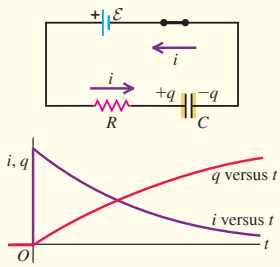


***R-C* circuits:** When a capacitor is charged by a battery in series with a resistor, the current and capacitor charge are not constant. The charge approaches its final value asymptotically and the current approaches zero asymptotically. The charge and current in the circuit are given by Eqs. (26.12) and (26.13). After a time $\tau = RC$, the charge has approached within $1/e$ of its final value. This time is called the time constant or relaxation time of the circuit. When the capacitor discharges, the charge and current are given as functions of time by Eqs. (26.16) and (26.17). The time constant is the same for charging and discharging. (See Examples 26.12 and 26.13.)

Capacitor charging:

$$q = C\mathcal{E}\left(1 - e^{-t/RC}\right) = Q_f\left(1 - e^{-t/RC}\right) \tag{26.12}$$

$$i = \frac{dq}{dt} = \frac{\mathcal{E}}{R}e^{-t/RC} = I_0 e^{-t/RC} \tag{26.13}$$

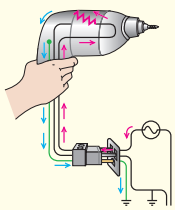


Capacitor discharging:

$$q = Q_0 e^{-t/RC} \tag{26.16}$$

$$i = \frac{dq}{dt} = -\frac{Q_0}{RC}e^{-t/RC} = I_0 e^{-t/RC} \tag{26.17}$$

Household wiring: In household wiring systems, the various electrical devices are connected in parallel across the supply. A mains electricity cable contains a pair of conductors, one "live" and the other "neutral." An additional "earth" wire is included for safety. The maximum permissible current in a circuit is determined by the size of the wires and the maximum temperature they can tolerate. Protection against excessive current and the resulting fire hazard is provided by fuses or circuit breakers. (See Example 26.14.)



Chapter 26 Media Assets

For assigned homework and other learning materials, go to Mastering Physics.

GUIDED PRACTICE

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLES 26.1 and 26.2 (Section 26.1) before attempting these problems.

VP26.2.1 You have three resistors: $R_1 = 1.00\,\Omega$, $R_2 = 2.00\,\Omega$, and $R_3 = 4.00\,\Omega$. Find the equivalent resistance for the combinations shown in (a) Fig. 26.1a, (b) Fig. 26.1b, (c) Fig. 26.1c, and (d) Fig. 26.1d.

VP26.2.2 You have three resistors, $R_1 = 4.00\,\Omega$, $R_2 = 5.00\,\Omega$, and $R_3 = 6.00\,\Omega$, connected as shown in Fig. 26.1d. The terminals of a battery with emf 24.0 V and negligible internal resistance are connected to points a and b. Find the current through (a) the battery, (b) R_1, (c) R_2, and (d) R_3.

VP26.2.3 You have three resistors, $R_1 = 7.00\,\Omega$, $R_2 = 8.00\,\Omega$, and $R_3 = 9.00\,\Omega$, connected as shown in Fig. 26.1b. The terminals of a battery with emf 12.0 V and negligible internal resistance are connected to points a and b. Find (a) the power output of the battery and the power input to (b) R_1, (c) R_2, and (d) R_3.

VP26.2.4 You have three resistors, $R_1 = 5.00\,\Omega$, $R_2 = 6.00\,\Omega$, and $R_3 = 7.00\,\Omega$, connected as shown in Fig. 26.1c. The terminals of a battery with emf 9.00 V and negligible internal resistance are connected to points a and b. Find (a) the power output of the battery and the power input to (b) R_1, (c) R_2, and (d) R_3.

Be sure to review EXAMPLES 26.3, 26.4, 26.5, 26.6, and 26.7 (Section 26.2) before attempting these problems.

VP26.7.1 In the circuit shown in Fig. 26.10a, you reverse the 4 V battery so that its positive terminal is on the right side instead of the left. Find (a) the current in the circuit, (b) the potential difference V_{ab}, and (c) the power output of the emf of each battery.

VP26.7.2 Figure 26.6a shows two batteries, one with emf $\mathcal{E}_1$ and one with emf $\mathcal{E}_2$, connected to a resistance R. Current I_1 flows through emf $\mathcal{E}_1$ from point c toward point a, and current I_2 flows through emf $\mathcal{E}_2$ from point b toward point a. If $\mathcal{E}_1 = 8.00$ V, $\mathcal{E}_2 = 9.00$ V, $R = 5.00\,\Omega$, $I_1 = 0.200$ A, and $I_2 = 1.35$ A, find the values of (a) the potential difference V_{ab}, (b) the internal resistance r_1, and (c) the internal resistance r_2.

VP26.7.3 In the circuit shown in Figure 26.12, you replace the 2 Ω resistor between points b and d with a 3 Ω resistor. Find the power dissipated in (a) the 1 Ω resistor between points c and a, (b) the 1 Ω resistor between points c and b, (c) the 1 Ω resistor between points a and b, (d) the 1 Ω resistor between points a and d, and (e) the 3 Ω resistor between points b and d. Give each answer to three significant figures.

VP26.7.4 In the circuit shown in Fig. 26.9a, let $r_1 = r_2 = 0$ (that is, the internal resistances are very small) and let $R_1 = R_2 = R_3 = R$. Find the values of (a) I_1, (b) I_2, and (c) I_3.

Be sure to review EXAMPLES 26.12 and 26.13 (Section 26.4) before attempting these problems.

VP26.13.1 You connect a 10.0 MΩ resistor in series with a 3.20 μF capacitor and a battery with emf 9.00 V. Before you close the switch at $t = 0$ to complete the circuit, the capacitor is uncharged. Find (a) the final capacitor charge, (b) the initial current, (c) the time constant, (d) the fraction of the final charge on the capacitor at $t = 18.0$ s, and (e) the fraction of the initial current present at $t = 18.0$ s.

VP26.13.2 A 2.20 μF capacitor initially has charge 4.20 μC. You connect it in series with a 4.00 MΩ resistor and an open switch. If $t = 0$ is the time when you close the switch, (a) at what time will the capacitor charge be 1.20 μC and (b) what will be the current in the circuit at this time?

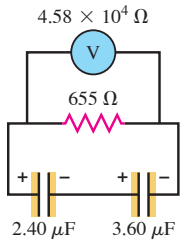
VP26.13.3 You connect an unknown resistor in series with an 8.00 μF capacitor and an open switch. Before you close the switch at $t = 0$, the capacitor has charge 5.50 μC; at $t = 17.0$ s, the charge has decreased to 1.10 μC. Find (a) the resistance of the resistor, (b) the current in the circuit just after you close the switch, and (c) the current in the circuit at $t = 17.0$ s.

VP26.13.4 You connect an initially uncharged 6.40 μF capacitor in series with a 5.00 MΩ resistor and a battery with emf 12.0 V. After letting the capacitor charge for 51.0 s, you disconnect it from this circuit and connect it in series to an open switch and a 6.00 MΩ resistor. Find the charge on the capacitor (a) when you disconnect it from the first circuit and (b) 70.0 s after you close the switch in the second circuit.

BRIDGING PROBLEM Two Capacitors and Two Resistors

A 2.40 μF capacitor and a 3.60 μF capacitor are connected in series. (a) A charge of 5.20 mC is placed on each capacitor. What is the energy stored in the capacitors? (b) A 655 Ω resistor is connected to the terminals of the capacitor combination, and a voltmeter with resistance $4.58 \times 10^4\ \Omega$ is connected across the resistor (**Fig. 26.27**). What is the rate of change of the energy stored in the capacitors just after the connection is made? (c) How long after the connection is made has the energy stored in the capacitors decreased to $1/e$ of its initial value? (d) At the instant calculated in part (c), what is the rate of change of the energy stored in the capacitors?

Figure **26.27** When the connection is made, the charged capacitors discharge.



SOLUTION GUIDE

IDENTIFY and SET UP

1. The two capacitors act as a single equivalent capacitor (see Section 24.2), and the resistor and voltmeter act as a single equivalent resistor. Select equations that will allow you to calculate the values of these equivalent circuit elements.
2. In part (a) you'll need to use Eq. (24.9), which gives the energy stored in a capacitor.
3. For parts (b), (c), and (d), you'll need to use Eq. (24.9) as well as Eqs. (26.16) and (26.17), which give the capacitor charge and current as functions of time. (*Hint:* The rate at which energy is lost by the capacitors equals the rate at which energy is dissipated in the resistances.)

EXECUTE

4. Find the stored energy at $t = 0$.
5. Find the rate of change of the stored energy at $t = 0$.
6. Find the value of t at which the stored energy has $1/e$ of the value you found in step 4.
7. Find the rate of change of the stored energy at the time you found in step 6.

EVALUATE

8. Check your results from steps 5 and 7 by calculating the rate of change in a different way. (*Hint:* The rate of change of the stored energy U is dU/dt.)

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

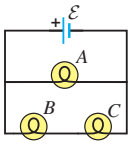
Q26.1 In which 230 V light bulb does the filament have greater resistance: a 60 W bulb or a 100 W bulb? If the two bulbs are connected to a 230 V supply in series, through which bulb will there be the greater voltage drop? What if they are connected in parallel? Explain your reasoning.

Q26.2 Two 120 V light bulbs, one 25 W and one 200 W, were connected in series across a 230 V supply. It seemed like a good idea at the time, but one bulb burned out almost immediately. Which one burned out, and why?

Q26.3 You connect a number of identical light bulbs to a flashlight battery. (a) What happens to the brightness of each bulb as more and more bulbs are added to the circuit if you connect them (i) in series and (ii) in parallel? (b) Will the battery last longer if the bulbs are in series or in parallel? Explain your reasoning.

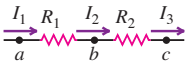
Q26.4 In the circuit shown in **Fig. Q26.4**, three identical light bulbs are connected to a flashlight battery. How do the brightnesses of the bulbs compare? Which light bulb has the greatest current passing through it? Which light bulb has the greatest potential difference between its terminals? What happens if bulb A is unscrewed? Bulb B? Bulb C? Explain your reasoning.

Figure **Q26.4**



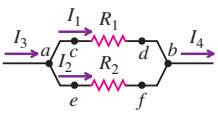
Q26.5 If two resistors R_1 and R_2 $(R_2 > R_1)$ are connected in series as shown in **Fig. Q26.5**, which of the following must be true? In each case justify your answer. (a) $I_1 = I_2 = I_3$. (b) The current is greater in R_1 than in R_2. (c) The electrical power consumption is the same for both resistors. (d) The electrical power consumption is greater in R_2 than in R_1. (e) The potential drop is the same across both resistors. (f) The potential at point a is the same as at point c. (g) The potential at point b is lower than at point c. (h) The potential at point c is lower than at point b.

Figure **Q26.5**



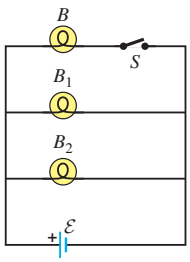
Q26.6 If two resistors R_1 and R_2 $(R_2 > R_1)$ are connected in parallel as shown in **Fig. Q26.6**, which of the following must be true? In each case justify your answer. (a) $I_1 = I_2$. (b) $I_3 = I_4$. (c) The current is greater in R_1 than in R_2. (d) The rate of electrical energy consumption is the same for both resistors. (e) The rate of electrical energy consumption is greater in R_2 than in R_1. (f) $V_{cd} = V_{ef} = V_{ab}$. (g) Point c is at higher potential than point d. (h) Point f is at higher potential than point e. (i) Point c is at higher potential than point e.

Figure **Q26.6**



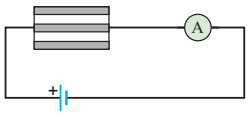
Q26.7 A battery with no internal resistance is connected across identical light bulbs as shown in **Fig. Q26.7**. When you close the switch S, will the brightness of bulbs B_1 and B_2 change? If so, how will it change? Explain.

Figure **Q26.7**



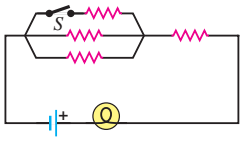
Q26.8 A resistor consists of three identical metal strips connected as shown in **Fig. Q26.8**. If one of the strips is cut out, does the ammeter reading increase, decrease, or stay the same? Why?

Figure **Q26.8**



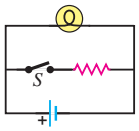
Q26.9 A light bulb is connected in the circuit shown in **Fig. Q26.9**. If we close the switch S, does the bulb's brightness increase, decrease, or remain the same? Explain why.

Figure **Q26.9**



Q26.10 A real battery, having nonnegligible internal resistance, is connected across a light bulb as shown in **Fig. Q26.10**. When the switch S is closed, what happens to the brightness of the bulb? Why?

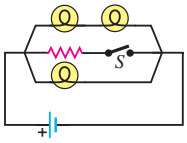
Figure **Q26.10**



Q26.11 If the battery in Discussion Question Q26.10 is ideal with no internal resistance, what will happen to the brightness of the bulb when S is closed? Why?

Q26.12 Consider the circuit shown in **Fig. Q26.12**. What happens to the brightnesses of the bulbs when the switch S is closed if the battery (a) has no internal resistance and (b) has nonnegligible internal resistance? Explain why.

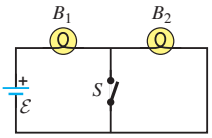
Figure **Q26.12**



Q26.13 Is it possible to connect resistors together in a way that cannot be reduced to some combination of series and parallel combinations? If so, give examples. If not, state why not.

Q26.14 The battery in the circuit shown in **Fig. Q26.14** has no internal resistance. After you close the switch S, will the brightness of bulb B_1 increase, decrease, or stay the same?

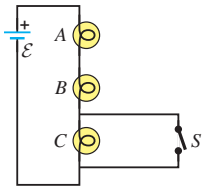
Figure **Q26.14**



Q26.15 In a two-cell flashlight, the batteries are usually connected in series. Why not connect them in parallel? What possible advantage could there be in connecting several identical batteries in parallel?

Q26.16 Identical light bulbs A, B, and C are connected as shown in **Fig. Q26.16**. When the switch S is closed, bulb C goes out. Explain why. What happens to the brightness of bulbs A and B? Explain.

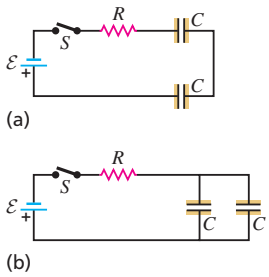
Figure **Q26.16**



Q26.17 The emf of a flashlight battery is roughly constant with time, but its internal resistance increases with age and use. What sort of meter should be used to test the freshness of a battery?

Q26.18 Will the capacitors in the circuits shown in **Fig. Q26.18** charge at the same rate when the switch S is closed? If not, in which circuit will the capacitors charge more rapidly? Explain.

Figure **Q26.18**



Q26.19 Verify that the time constant RC has units of time.

Q26.20 For very large resistances it is easy to construct R-C circuits that have time constants of several seconds or minutes. How might this fact be used to measure very large resistances, those that are too large to measure by more conventional means?

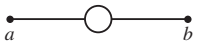
Q26.21 When a capacitor, battery, and resistor are connected in series, does the resistor affect the maximum charge stored on the capacitor? Why or why not? What purpose does the resistor serve?

EXERCISES

Section 26.1 Resistors in Series and Parallel

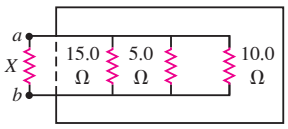
26.1 •• A uniform wire of resistance R is cut into three equal lengths. One of these is formed into a circle and connected between the other two (**Fig. E26.1**). What is the resistance between the opposite ends a and b?

Figure **E26.1**



26.2 •• A machine part has a resistor X protruding from an opening in the side. This resistor is connected to three other resistors, as shown in **Fig. E26.2**. An ohmmeter connected across a and b reads 2.00 Ω. What is the resistance of X?

Figure **E26.2**

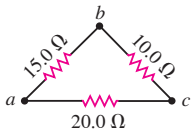


26.3 • A resistor with $R_1 = 25.0\ \Omega$ is connected to a battery that has negligible internal resistance and electrical energy is dissipated by R_1 at a rate of 36.0 W. If a second resistor with $R_2 = 15.0\ \Omega$ is connected in series with R_1, what is the total rate at which electrical energy is dissipated by the two resistors?

26.4 • A 46 Ω resistor and a 20 Ω resistor are connected in parallel, and the combination is connected across a 230 V dc supply. (a) What is the resistance of the parallel combination? (b) What is the total current through the parallel combination? (c) What is the current through each resistor?

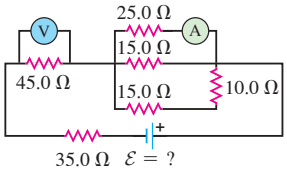
26.5 • A triangular array of resistors is shown in **Fig. E26.5**. What current will this array draw from a 30.0 V battery having negligible internal resistance if we connect it across (a) ab; (b) bc; (c) ac? (d) If the battery has an internal resistance of 5.00 Ω, what current will the array draw if the battery is connected across bc?

Figure **E26.5**



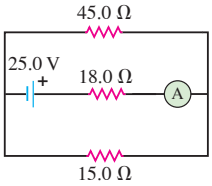
26.6 •• For the circuit shown in **Fig. E26.6** both meters are idealized, the battery has no appreciable internal resistance, and the ammeter reads 1.15 A. (a) What does the voltmeter read? (b) What is the emf $\mathcal{E}$ of the battery?

Figure **E26.6**



26.7 •• For the circuit shown in **Fig. E26.7** find the reading of the idealized ammeter if the battery has an internal resistance of 3.86 Ω.

Figure **E26.7**



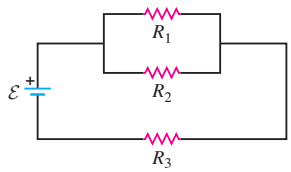
26.8 • Two resistors, R_1 and R_2, are connected in parallel to a power supply that has voltage V and negligible internal resistance. $R_2 = 8.00\ \Omega$ and the resistance of R_1 is not known. For several values of V, you measure the current I flowing through the voltage source. You plot the data as I versus V and find that they lie close to a straight line that has slope $0.208\ \Omega^{-1}$. What is the resistance of R_1?

26.9 • Six identical resistors, each with resistance R, are connected to an emf $\mathcal{E}$. (a) In terms of $\mathcal{E}$ and R, what is the current I through each of the resistors if they are connected in parallel? (b) In series? (c) For which network of resistors, series or parallel, is the power consumed in each resistor greater?

26.10 •• **Power Rating of a Resistor.** The *power rating* of a resistor is the maximum power the resistor can safely dissipate without too great a rise in temperature and hence damage to the resistor. (a) If the power rating of a 15 kΩ resistor is 5.0 W, what is the maximum allowable potential difference across the terminals of the resistor? (b) A 10.0 kΩ resistor is to be connected across a 230 V potential difference. What power rating is required? (c) A 100.0 Ω and a 150.0 Ω resistor, both rated at 3.00 W, are connected in series across a variable potential difference. What is the greatest this potential difference can be without overheating either resistor, and what is the rate of heat generated in each resistor under these conditions?

26.11 • In **Fig. E26.11**, $R_1 = 6.00\ \Omega$, $R_2 = 6.00\ \Omega$, and $R_3 = 7.00\ \Omega$. The battery has negligible internal resistance. The current I_2 through R_2 is 4.00 A. (a) What are the currents I_1 and I_3? (b) What is the emf of the battery?

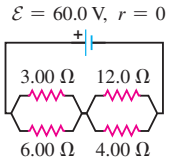
Figure **E26.11**



26.12 •• In Fig. E26.11 the battery has emf 35.0 V and negligible internal resistance. $R_1 = 5.00\ \Omega$. The current through R_1 is 1.50 A, and the current through $R_3 = 4.50$ A. What are the resistances R_2 and R_3?

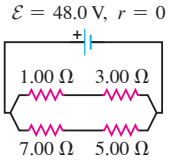
26.13 • Compute the equivalent resistance of the network in **Fig. E26.13**, and find the current in each resistor. The battery has negligible internal resistance.

Figure **E26.13**



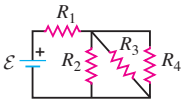
26.14 • Compute the equivalent resistance of the network in **Fig. E26.14**, and find the current in each resistor. The battery has negligible internal resistance.

Figure **E26.14**



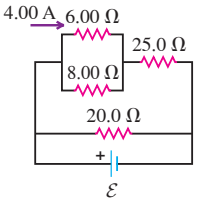
26.15 • In the circuit of **Fig. E26.15**, each resistor represents a light bulb. Let $R_1 = R_2 = R_3 = R_4 = 4.37\ \Omega$ and $\mathcal{E} = 9.02$ V. (a) Find the current in each bulb. (b) Find the power dissipated in each bulb. Which bulb or bulbs glow the brightest? (c) Bulb R_4 is now removed from the circuit, leaving a break in the wire at its position. Now what is the current in each of the remaining bulbs R_1, R_2, and R_3? (d) With bulb R_4 removed, what is the power dissipated in each of the remaining bulbs? (e) Which light bulb(s) glow brighter as a result of removing R_4? Which bulb(s) glow less brightly? Discuss why there are different effects on different bulbs.

Figure **E26.15**



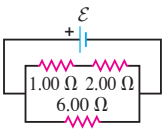
26.16 • Consider the circuit shown in **Fig. E26.16**. The current through the 6.00 Ω resistor is 4.00 A, in the direction shown. What are the currents through the 25.0 Ω and 20.0 Ω resistors?

Figure **E26.16**



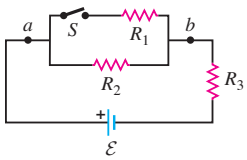
26.17 • In the circuit shown in **Fig. E26.17**, the voltage across the 2.00 Ω resistor is 12.0 V. What are the emf of the battery and the current through the 6.00 Ω resistor?

Figure **E26.17**



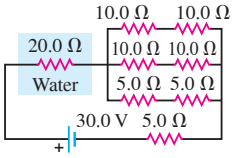
26.18 •• In the circuit shown in **Fig. E26.18**, $\mathcal{E} = 36.0$ V, $R_1 = 4.00\ \Omega$, $R_2 = 6.00\ \Omega$, and $R_3 = 3.00\ \Omega$. (a) What is the potential difference V_{ab} between points a and b when the switch S is open and when S is closed? (b) For each resistor, calculate the current through the resistor with S open and with S closed. For each resistor, does the current increase or decrease when S is closed?

Figure **E26.18**



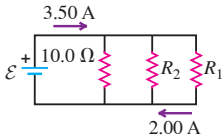
26.19 •• **CP** In the circuit in **Fig. E26.19**, a 20.0 Ω resistor is inside 100 g of pure water that is surrounded by insulating styrofoam. If the water is initially at 10.0°C, how long will it take for its temperature to rise to 57.0°C?

Figure **E26.19**



26.20 • In the circuit shown in **Fig. E26.20**, the rate at which R_1 is dissipating electrical energy is 15.0 W. (a) Find R_1 and R_2. (b) What is the emf of the battery? (c) Find the current through both R_2 and the 10.0 Ω resistor. (d) Calculate the total electrical power consumption in all the resistors and the electrical power delivered by the battery. Show that your results are consistent with conservation of energy.

Figure **E26.20**

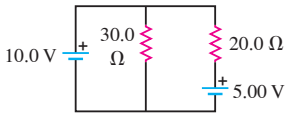


26.21 • **Light Bulbs in Series and in Parallel.** Two light bulbs have constant resistances of 40 Ω and 80 Ω. If the two light bulbs are connected in series across a 12 V supply, find (a) the current through each bulb; (b) the power dissipated in each bulb; (c) the total power dissipated in both bulbs. The two light bulbs are now connected in parallel across the 12 V supply. Find (d) the current through each bulb; (e) the power dissipated in each bulb; (f) the total power dissipated in both bulbs. (g) In each situation, which of the two bulbs glows the brightest? (h) In which situation is there a greater total light output from both bulbs combined?

Section 26.2 Kirchhoff's Rules

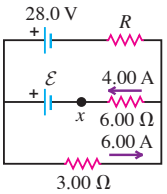
26.22 •• The batteries shown in the circuit in **Fig. E26.22** have negligibly small internal resistances. Find the current through (a) the 30.0 Ω resistor; (b) the 20.0 Ω resistor; (c) the 10.0 V battery.

Figure **E26.22**



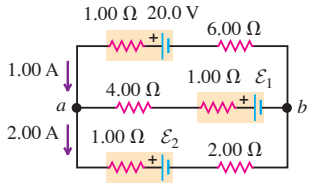
26.23 • In the circuit shown in **Fig. E26.23** find (a) the current in resistor R; (b) the resistance R; (c) the unknown emf $\mathcal{E}$. (d) If the circuit is broken at point x, what is the current in resistor R?

Figure **E26.23**



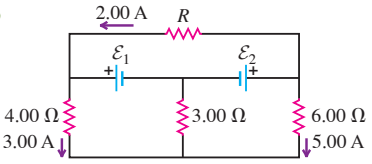
26.24 • Find the emfs $\mathcal{E}_1$ and $\mathcal{E}_2$ in the circuit of **Fig. E26.24**, and find the potential difference of point b relative to point a.

Figure E26.24



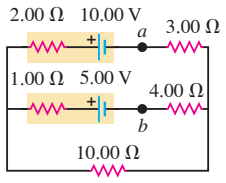
26.25 • In the circuit shown in **Fig. E26.25**, find (a) the current in the 3.00 Ω resistor; (b) the unknown emfs $\mathcal{E}_1$ and $\mathcal{E}_2$; (c) the resistance R. Note that three currents are given.

Figure E26.25



26.26 •• In the circuit shown in **Fig. E26.26**, find (a) the current in each branch and (b) the potential difference V_{ab} of point a relative to point b.

Figure E26.26

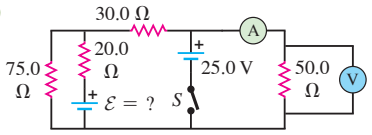


26.27 • The 10.00 V battery in Fig. E26.26 is removed from the circuit and reinserted with the opposite polarity, so that its positive terminal is now next to point a. The rest of the circuit is as shown in the figure. Find (a) the current in each branch and (b) the potential difference V_{ab} of point a relative to point b.

26.28 • The 5.00 V battery in Fig. E26.26 is removed from the circuit and replaced by a 15.00 V battery, with its negative terminal next to point b. The rest of the circuit is as shown in the figure. Find (a) the current in each branch and (b) the potential difference V_{ab} of point a relative to point b.

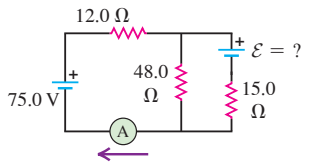
26.29 •• In the circuit shown in **Fig. E26.29** the batteries have negligible internal resistance and the meters are both idealized. With the switch S open, the voltmeter reads 17.0 V. (a) Find the emf $\mathcal{E}$ of the battery. (b) What will the ammeter read when the switch is closed?

Figure E26.29



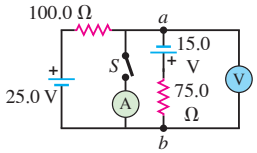
26.30 •• In the circuit shown in **Fig. E26.30** both batteries have insignificant internal resistance and the idealized ammeter reads 1.50 A in the direction shown. Find the emf $\mathcal{E}$ of the battery. Is the polarity shown correct?

Figure E26.30



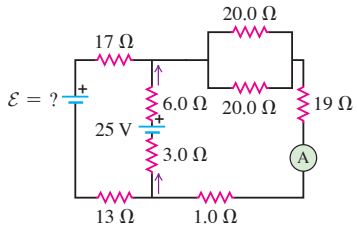
26.31 • In the circuit shown in **Fig. E26.31** all meters are idealized and the batteries have no appreciable internal resistance. (a) Find the reading of the voltmeter with the switch S open. Which point is at a higher potential: a or b? (b) With S closed, find the reading of the voltmeter and the ammeter. Which way (up or down) does the current flow through the switch?

Figure E26.31



26.32 •• In the circuit shown in **Fig. E26.32**, the 6.0 Ω resistor is consuming energy at a rate of 23 J/s when the current through it flows as shown. (a) Find the current through the ammeter A. (b) What are the polarity and emf $\mathcal{E}$ of the unknown battery, assuming it has negligible internal resistance?

Figure E26.32

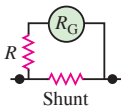


Section 26.3 Electrical Measuring Instruments

26.33 • The resistance of a galvanometer coil is 25.0 Ω, and the current required for full-scale deflection is 500 μA. (a) Show in a diagram how to convert the galvanometer to an ammeter reading 20.0 mA full scale, and compute the shunt resistance. (b) Show how to convert the galvanometer to a voltmeter reading 500 mV full scale, and compute the series resistance.

26.34 • The resistance of the coil of a pivoted-coil galvanometer is 9.36 Ω, and a current of 0.0224 A causes it to deflect full scale. We want to convert this galvanometer to an ammeter reading 20.0 A full scale. The only shunt available has a resistance of 0.0250 Ω. What resistance R must be connected in series with the coil (**Fig. E26.34**)?

Figure E26.34



26.35 • A circuit consists of a series combination of 5.50-kΩ and 4.50-kΩ resistors connected across a 50.0-V battery having negligible internal resistance. You want to measure the true potential difference (that is, the potential difference without the meter present) across the 4.50-kΩ resistor using a voltmeter having an internal resistance of 10.0 kΩ. (a) What potential difference does the voltmeter measure across the 4.50-kΩ resistor? (b) What is the *true* potential difference across this resistor when the meter is not present? (c) By what percentage is the voltmeter reading in error from the true potential difference?

Section 26.4 *R-C* Circuits

26.36 • To measure the capacitance C of a capacitor, you attach the capacitor to a battery and wait until it is fully charged. You then disconnect the capacitor from the battery and let it discharge through a resistor of resistance R. You measure the time $T_{1/2}$ that it takes the voltage across the resistor to decrease to half its initial value at the instant that the connection to the capacitor is first completed. You repeat this for several different resistors. You plot the data as $T_{1/2}$ versus R and find that they lie close to a straight line that has slope 5.00 μF. What is the capacitance C of the capacitor?

26.37 • A capacitor is charged to a potential of 12.0 V and is then connected to a voltmeter having a total internal resistance of 3.40 MΩ. After a time of 4.00 s the voltmeter reads 3.0 V. What are (a) the capacitance and (b) the time constant of the circuit?

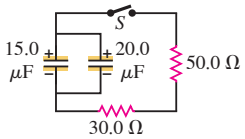
26.38 •• You connect a battery, resistor, and capacitor as in Fig. 26.20a, where $\mathcal{E} = 36.0$ V, $C = 5.00\ \mu$F, and $R = 120\ \Omega$. The switch S is closed at $t = 0$. (a) When the voltage across the capacitor is 8.00 V, what is the magnitude of the current in the circuit? (b) At what time t after the switch is closed is the voltage across the capacitor 8.00 V? (c) When the voltage across the capacitor is 8.00 V, at what rate is energy being stored in the capacitor?

26.39 • A 4.60 μF capacitor that is initially uncharged is connected in series with a 7.50 kΩ resistor and an emf source with $\mathcal{E} = 245$ V and negligible internal resistance. Just after the circuit is completed, what are (a) the voltage drop across the capacitor; (b) the voltage drop across the resistor; (c) the charge on the capacitor; (d) the current through the resistor? (e) A long time after the circuit is completed (after many time constants) what are the values of the quantities in parts (a)–(d)?

26.40 •• You connect a battery, resistor, and capacitor as in Fig. 26.20a, where $R = 12.0\ \Omega$ and $C = 5.00 \times 10^{-6}$ F. The switch S is closed at $t = 0$. When the current in the circuit has magnitude 3.00 A, the charge on the capacitor is 40.0×10^{-6} C. (a) What is the emf of the battery? (b) At what time t after the switch is closed is the charge on the capacitor equal to 40.0×10^{-6} C? (c) When the current has magnitude 3.00 A, at what rate is energy being (i) stored in the capacitor, (ii) supplied by the battery?

26.41 •• CP In the circuit shown in **Fig. E26.41** both capacitors are initially charged to 50.0 V. (a) How long after closing the switch S will the potential across each capacitor be reduced to 15.0 V, and (b) what will be the current at that time?

Figure E26.41



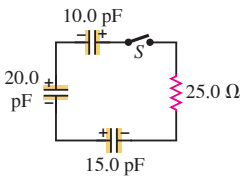
26.42 • A 12.6-μF capacitor is connected through a 0.895-MΩ resistor to a constant potential difference of 60.0 V. (a) Compute the charge on the capacitor at the following times after the connections are made: 0, 5.0 s, 10.0 s, 20.0 s, and 100.0 s. (b) Compute the charging currents at the same instants. (c) Graph the results of parts (a) and (b) for t between 0 and 20 s.

26.43 • An emf source with $\mathcal{E} = 230$ V, a resistor with $R = 120.0\ \Omega$, and a capacitor with $C = 4.00\ \mu$F are connected in series. As the capacitor charges, when the current in the resistor is 1.150 A, what is the magnitude of the charge on each plate of the capacitor?

26.44 • A network of two identical capacitors, each with capacitance C, is charged through a resistor R using a battery with emf $\mathcal{E}$. (a) What is the time constant, in terms of R and C, for the charging circuit if the two capacitors are in series? (b) In parallel? (c) For which capacitor network, series or parallel, does the voltage across the resistor become 1% of its initial value in a shorter time?

26.45 •• CP In the circuit shown in **Fig. E26.45** each capacitor initially has a charge of magnitude 3.50 nC on its plates. After the switch S is closed, what will be the current in the circuit at the instant that the capacitors have lost 80.0% of their initial stored energy?

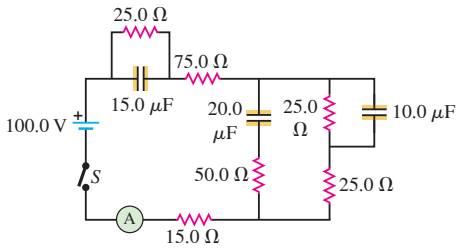
Figure E26.45



26.46 • A 1.60-μF capacitor is charging through a 14.0-Ω resistor using a 10.0-V battery. What will be the current when the capacitor has acquired $\frac{1}{4}$ of its maximum charge? Will it be $\frac{1}{4}$ of the maximum current?

26.47 • In the circuit in **Fig. E26.47** the capacitors are initially uncharged, the battery has no internal resistance, and the ammeter is idealized. Find the ammeter reading (a) just after the switch S is closed and (b) after S has been closed for a very long time.

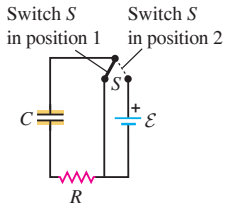
Figure E26.47



26.48 • A 12.0 μF capacitor is charged to a potential of 50.0 V and then discharged through a 225 Ω resistor. How long does it take the capacitor to lose (a) half of its charge and (b) half of its stored energy?

26.49 • In the circuit shown in **Fig. E26.49**, $C = 5.90\ \mu$F, $\mathcal{E} = 28.0$ V, and the emf has negligible resistance. Initially the capacitor is uncharged and the switch S is in position 1. The switch is then moved to position 2, so that the capacitor begins to charge. (a) What will be the charge on the capacitor a long time after S is moved to position 2? (b) After S has been in position 2 for 3.00 ms, the charge on the capacitor is measured to be 110 μC. What is the value of the resistance R? (c) How long after S is moved to position 2 will the charge on the capacitor be equal to 99.0% of the final value found in part (a)?

Figure E26.49



26.50 • A capacitor with $C = 6.00\ \mu$F is fully charged by connecting it to a battery that has emf 50.0 V. The capacitor is disconnected from the battery. A resistor of resistance $R = 185\ \Omega$ is connected across the capacitor, and the capacitor discharges through the resistor. (a) What is the charge q on the capacitor when the current in the resistor is 0.180 A? (b) If the connection to the resistor is completed at time $t = 0$, what is the value of t when the current has the value specified in part (a)?

Section 26.5 Power Distribution Systems

26.51 • A 2000 W electric heater is plugged into the outlet of a 230 V circuit that has a 16 A circuit breaker. You plug an electric hairdryer into the same outlet. The hairdryer has power settings of 1500 W, 1800 W, 2100 W, and 2400 W. You start with the hairdryer on the 1500 W setting and increase the power setting until the circuit breaker trips. What power setting caused the breaker to trip?

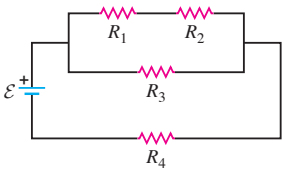
26.52 • The heating element of tumble dryer is rated at 3.5 kW when connected to a 230 V supply. (a) What is the current in the heating element? Is 2.5 mm^2 section wire large enough to supply this current? (b) What is the resistance of the dryer's heating element at its operating temperature? (c) At £ 0.125 per kWh, how much does it cost per hour to operate the dryer?

PROBLEMS

26.53 •• **CALC** A capacitor with capacitance C is connected in series to a resistor of resistance R and a battery with emf $\mathcal{E}$. The circuit is completed at time $t = 0$. (a) In terms of $\mathcal{E}$, R, and C, how much energy is stored in the capacitor when it is fully charged? (b) The power output of the battery is $P_{\mathcal{E}} = \mathcal{E}i$, with i given by Eq. (26.13). The electrical energy supplied in an infinitesimal time interval dt is $P_{\mathcal{E}}dt$. Integrate from $t = 0$ to $t \to \infty$ to find the total energy supplied by the battery. (c) The rate of consumption of electrical energy in the resistor is $P_R = i^2R$. In an infinitesimal time interval dt, the amount of electrical energy consumed by the resistor is $P_R dt$. Integrate from $t = 0$ to $t \to \infty$ to find the total energy consumed by the resistor. (d) What fraction of the total energy supplied by the battery is stored in the capacitor? What fraction is consumed in the resistor?

26.54 •• In **Fig. P26.54**, the battery has negligible internal resistance and $\mathcal{E} = 48.0$ V. $R_1 = R_2 = 3.70\ \Omega$ and $R_4 = 4.00\ \Omega$. What must the resistance R_3 be for the resistor network to dissipate electrical energy at a rate of 270 W?

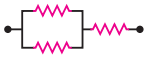
Figure **P26.54**



26.55 •• Consider the circuit shown in Fig. 26.6a. $\mathcal{E}_1 = 24.0$ V, $\mathcal{E}_2 = 36.0$ V, $r_1 = r_2 = 2.00\ \Omega$, and $R = 20.0\ \Omega$. (a) What is the current I_1 through $\mathcal{E}_1$? (b) What is the current I_2 through $\mathcal{E}_2$? (c) What is the current I_R through R?

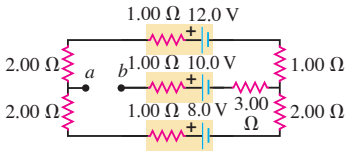
26.56 •• Each of the three resistors in **Fig. P26.56** has a resistance of 3.5 Ω and can dissipate a maximum of 34 W without becoming excessively heated. What is the maximum power the circuit can dissipate?

Figure **P26.56**



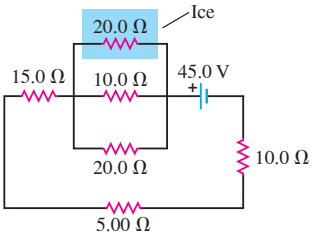
26.57 •• (a) Find the potential of point a with respect to point b in Fig. P26.57. (b) If points a and b are connected by a wire with negligible resistance, find the current in the 8.0-V battery.

Figure **P26.57**



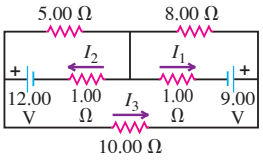
26.58 •• **CP** For the circuit shown in **Fig. P26.58** a 20.0 Ω resistor is embedded in a large block of ice at 0.00°C, and the battery has negligible internal resistance. At what rate (in g/s) is this circuit melting the ice? (The latent heat of fusion for ice is 3.34×10^5 J/kg.)

Figure **P26.58**



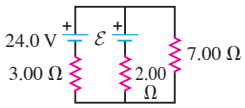
26.59 • Calculate the three currents I_1, I_2, and I_3 indicated in the circuit diagram shown in **Fig. P26.59**.

Figure **P26.59**



26.60 ••• What must the emf $\mathcal{E}$ in **Fig. P26.60** be in order for the current through the 7.00 Ω resistor to be 1.80 A? Each emf source has negligible internal resistance.

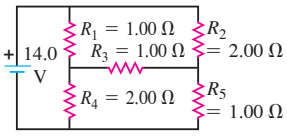
Figure **P26.60**



26.61 •• A standard phone charger provides 5.0 V of potential and supplies 1.0 A of current to the phone. (a) Estimate the diameter of each of the two wires inside your charge cable. (b) Assume the wire is copper, which has a resistivity of $1.72 \times 10^{-8}\ \Omega \cdot$m, and determine its resistance per meter. (c) If the phone draws $I = 1.0$ A, the potential supplied to the phone will be lower than the potential on the side of the charger by an amount IR, where R is the resistance of the wire. Determine the length of charging wire that would diminish the potential to 4.0 V at the phone.

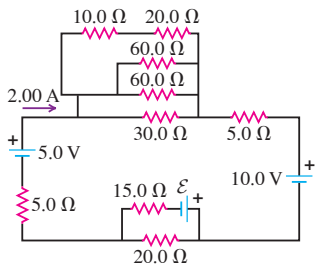
26.62 • (a) Find the current through the battery and each resistor in the circuit shown in **Fig. P26.62**. (b) What is the equivalent resistance of the resistor network?

Figure **P26.62**



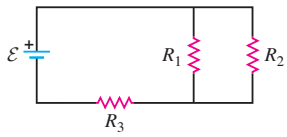
26.63 •• Consider the circuit shown in **Fig. P26.63**. (a) What must the emf $\mathcal{E}$ of the battery be in order for a current of 2.00 A to flow through the 5.00 V battery as shown? Is the polarity of the battery correct as shown? (b) How long does it take for 60.0 J of thermal energy to be produced in the 10.0 Ω resistor?

Figure **P26.63**



26.64 •• In the circuit shown in **Fig. P26.64**, $\mathcal{E} = 24.0$ V, $R_1 = 6.00\ \Omega$, $R_3 = 12.0\ \Omega$, and R_2 can vary between 3.00 Ω and 24.0 Ω. For what value of R_2 is the power dissipated by heating element R_1 the greatest? Calculate the magnitude of the greatest power.

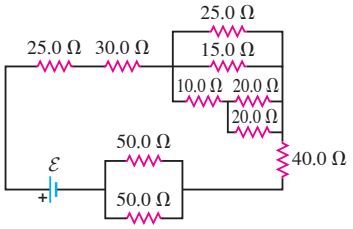
Figure **P26.64**



26.65 •• (a) Estimate energy usage and cost in your home by creating a table: In the first column list the household appliances that use electricity. In the second column indicate the number of daily hours T each appliance is used, and in the third column list the power P drawn by each appliance (This can be found on the appliance's label, in the technical data sheet provided by the manufacturer or on the Internet). In the fourth column compute the daily energy use $U = PT$, in kilowatt-hours. At the bottom of the final column, sum the total kilowatt-hours used by these appliances. (b) Compute the daily cost of electricity by multiplying the total daily kilowatt-hours by the cost in your city (for example, £ 0.125/kWh in the UK). Then multiply the result by 30 days to obtain an estimate of your monthly cost of electricity.

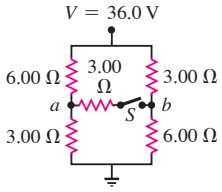
26.66 ••• In the circuit shown in **Fig. P26.66** all the resistors are rated at a maximum power of 2.00 W. What is the maximum emf $\mathcal{E}$ that the battery can have without burning up any of the resistors?

Figure **P26.66**



26.67 • **Figure P26.67** employs a convention often used in circuit diagrams. The battery (or other power supply) is not shown explicitly. It is understood that the point at the top, labeled "36.0 V," is connected to the positive terminal of a 36.0 V battery having negligible internal resistance, and that the *earth* symbol at the bottom is connected to the negative terminal of the battery. The circuit is completed through the battery, even though it is not shown. (a) What is the potential difference V_{ab}, the potential of point a relative to point b, when the switch S is open? (b) What is the current through S when it is closed? (c) What is the equivalent resistance when S is closed?

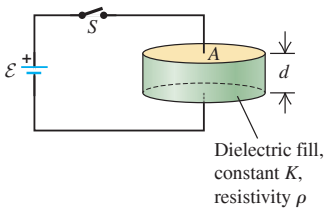
Figure **P26.67**



26.68 •• Three identical resistors are connected in series. When a certain potential difference is applied across the combination, the total power dissipated is 46.0 W. What power would be dissipated if the three resistors were connected in parallel across the same potential difference?

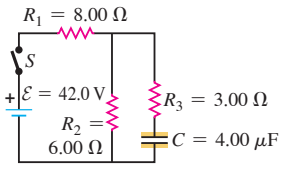
26.69 ••• **CP** Two capacitor plates with area A are separated by a distance d. The space between the plates is filled with dielectric material with dielectric constant K and resistivity ρ. This capacitor is attached to a battery that supplies a constant potential $\mathcal{E}$, as shown in **Fig. P26.69**, and is fully charged. At time $t = 0$ the switch S is opened. Owing to the nonzero value of ρ, this capacitor discharges by leaking. We can model this device as a capacitor and a resistor in parallel. (a) Analyze this circuit and determine the time constant, characterizing the discharge in terms of the parameters given above. Now assume the capacitor has plates with area $A = 1.00\ \text{cm}^2$ separated by $d = 30.0\ \mu\text{m}$ and is filled with a ceramic of dielectric constant $K = 8.70$ and resistivity $\rho = 3.10 \times 10^{12}\ \Omega \cdot \text{m}$. (b) If $\mathcal{E} = 5.00$ V, what is the charge on the capacitor at $t = 0$? (c) At what time will the capacitor have half of its original charge? (d) What is the magnitude of the leaking current at that point?

Figure **P26.69**



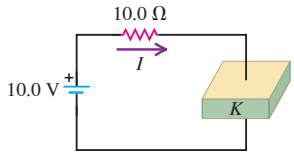
26.70 • The capacitor in **Fig. P26.70** is initially uncharged. The switch S is closed at $t = 0$. (a) Immediately after the switch is closed, what is the current through each resistor? (b) What is the final charge on the capacitor?

Figure **P26.70**



26.71 ••• **CP CALC** An R-C circuit is constructed as shown in **Fig. P26.71**. The capacitor has 2.00 cm × 2.00 cm square plates that are parallel and separated by a distance of 3.00 mm, and it is fully charged. At time $t = 0$, a 2.00 cm × 2.00 cm × 3.00 mm slab with dielectric constant $K = 12.0$ is slid between the plates as shown. (a) How much charge is on the upper plate before the slab is inserted? (b) What is the current I immediately after the slab is inserted? (c) How much energy is stored in the capacitor immediately after the slab is inserted? (d) By how much does the energy in the capacitor increase from that value after a long time has passed? (e) How much total energy is eventually supplied by the battery after the slab is inserted? (*Hint:* Determine the current as a function of time; then integrate the power delivered by the battery, $\int_0^\infty VI(t)\,dt$.) (f) How much energy is dissipated by the resistor?

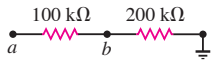
Figure **P26.71**



26.72 •• A 5.80-μF capacitor that is initially uncharged is connected in series with a 4.90-Ω resistor and an emf source with $\mathcal{E} = 53.0$ V and negligible internal resistance. At the instant when the resistor is dissipating electrical energy at a rate of 287 W, how much energy has been stored in the capacitor?

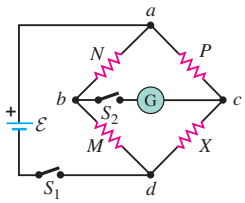
26.73 • Point a in **Fig. P26.73** is maintained at a constant potential of 400 V above earth potential. (See Problem 26.67.) (a) What is the reading of a voltmeter with the proper range and with resistance $5.00 \times 10^4\ \Omega$ when connected between point b and earth? (b) What is the reading of a voltmeter with resistance $5.00 \times 10^6\ \Omega$? (c) What is the reading of a voltmeter with infinite resistance?

Figure **P26.73**



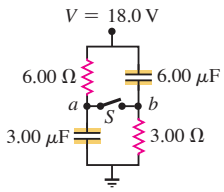
26.74 •• **The Wheatstone Bridge.** The circuit shown in **Fig. P26.74**, called a Wheatstone bridge, is used to determine the value of an unknown resistor X by comparison with three resistors M, N, and P whose resistances can be varied. For each setting, the resistance of each resistor is precisely known. With switches S_1 and S_2 closed, these resistors are varied until the current in the galvanometer G is zero; the bridge is then said to be *balanced*. (a) Show that under this condition the unknown resistance is given by $X = MP/N$. (This method permits very high precision in comparing resistors.) (b) If galvanometer G shows zero deflection when $M = 860\ \Omega$, $N = 14.00\ \Omega$, and $P = 33.38\ \Omega$, what is the unknown resistance X?

Figure **P26.74**



26.75 • (See Problem 26.67.) (a) What is the potential of point a with respect to point b in **Fig. P26.75** when the switch S is open? (b) Which point, a or b, is at the higher potential? (c) What is the final potential of point b with respect to earth when S is closed? (d) How much does the charge on each capacitor change when S is closed?

Figure **P26.75**

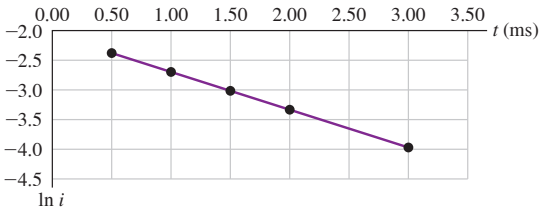


26.76 • A resistor with $R = 840\ \Omega$ is connected to the plates of a charged capacitor with capacitance $C = 4.58\ \mu$F. Just before the connection is made, the charge on the capacitor is 8.10 mC. (a) What is the energy initially stored in the capacitor? (b) What is the electrical power dissipated in the resistor just after the connection is made? (c) What is the electrical power dissipated in the resistor at the instant when the energy stored in the capacitor has decreased to half the value calculated in part (a)?

26.77 • A capacitor that is initially uncharged is connected in series with a resistor and an emf source with $\mathcal{E} = 110$ V and negligible internal resistance. Just after the circuit is completed, the current through the resistor is 6.5×10^{-5} A. The time constant for the circuit is 5.2 s. What are the resistance of the resistor and the capacitance of the capacitor?

26.78 •• **DATA** You set up the circuit shown in Fig. 26.22a, where $R = 196\ \Omega$. You close the switch at time $t = 0$ and measure the magnitude i of the current in the resistor R as a function of time t since the switch was closed. Your results are shown in **Fig. P26.78**, where you have chosen to plot $\ln i$ as a function of t. (a) Explain why your data points lie close to a straight line. (b) Use the graph in Fig. P26.78 to calculate the capacitance C and the initial charge Q_0 on the capacitor. (c) When $i = 0.0500$ A, what is the charge on the capacitor? (d) When $q = 0.500 \times 10^{-4}$ C, what is the current in the resistor?

Figure **P26.78**



26.79 •• **DATA** You set up the circuit shown in Fig. 26.20, where $C = 5.00 \times 10^{-6}$ F. At time $t = 0$, you close the switch and then measure the charge q on the capacitor as a function of the current i in the resistor. Your results are given in the table:

i (mA)	56.0	48.0	40.0	32.0	24.0
q (μC)	10.1	19.8	30.2	40.0	49.9

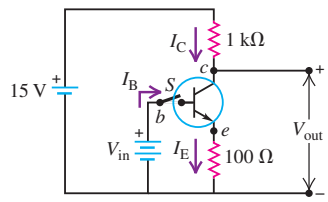
(a) Graph q as a function of i. Explain why the data points, when plotted this way, fall close to a straight line. Find the slope and y-intercept of the straight line that gives the best fit to the data. (b) Use your results from part (a) to calculate the resistance R of the resistor and the emf $\mathcal{E}$ of the battery. (c) At what time t after the switch is closed is the voltage across the capacitor equal to 10.0 V? (d) When the voltage across the capacitor is 4.00 V, what is the voltage across the resistor?

26.80 •• **DATA** The electronics supply company where you work has two different resistors, R_1 and R_2, in its inventory, and you must measure the values of their resistances. Unfortunately, stock is low, and all you have are R_1 and R_2 in parallel and in series—and you can't separate these two resistor combinations. You separately connect each resistor network to a battery with emf 48.0 V and negligible internal resistance and measure the power P supplied by the battery in both cases. For the series combination, $P = 48.0$ W; for the parallel combination, $P = 256$ W. You are told that $R_1 > R_2$. (a) Calculate R_1 and R_2. (b) For the series combination, which resistor consumes more power, or do they consume the same power? Explain. (c) For the parallel combination, which resistor consumes more power, or do they consume the same power?

CHALLENGE PROBLEMS

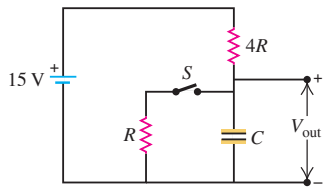
26.81 ••• The three-terminal device circled in **Fig. P26.81** is an NPN transistor. It works using three simple rules: I. The net current into the device is the same as the net current out of the device, so $I_C + I_B = I_E$. II. The potential at point e is always 0.60 V less than the potential at point b. Thus, $V_{in} - V_e = 0.60$ V. III. The current into point c is always a fixed multiple of the current into point b, so $I_C = \beta I_B$, where $\beta \gg 1$ is a parameter characteristic of the transistor. Use these rules to answer the questions. (a) What is the potential V_{out} in terms of V_{in}, in the limit $\beta \to \infty$? (*Hint:* Determine I_E in terms of V_{in}, and use this to determine I_C in terms of V_{in} and β. The potential difference across the 1.0 kΩ resistor can then be used to determine V_{out}. Take the limit $\beta \to \infty$ in this result.) (b) What value of V_{in} is needed so that $V_{out} = 7.5$ V? (c) If the input potential includes a small time-dependent "signal," so that $V_{in} = 15.0\text{ V} + v_{in}(t)$, then the output potential is $V_{out} = 7.5\text{ V} + Gv_{in}(t)$, where G is a "gain" factor. What is G for this circuit?

Figure **P26.81**



26.82 ••• In the circuit shown in **Fig. P26.82**, the switch S has been closed for a long time. (a) What is the output voltage V_{out}? Now the switch is opened and the output voltage increases. When the output voltage reaches 10.0 V, the switch closes and then the capacitor begins to discharge. When the output voltage reaches 5.0 V, the switch opens again. Thus this circuit undergoes periodic cycles of charging and discharging. (b) What is the time constant during a charging phase? (c) What is the time constant during a discharging phase? (d) What length of time passes between subsequent 10.0 V output voltages? (e) If $R = 10.0$ kΩ and $C = 10.0\ \mu$F, what is the frequency of operation? (Note: This circuit is a simplified example of a timer circuit, a crucial element in many electronics applications.)

Figure **P26.82**

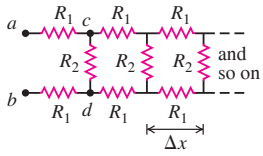


26.83 ••• **An Infinite Network.** As shown in **Fig. P26.83**, a network of resistors of resistances R_1 and R_2 extends to infinity toward the right. Prove that the total resistance R_T of the infinite network is equal to

$$R_T = R_1 + \sqrt{R_1^2 + 2R_1R_2}$$

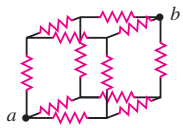
(*Hint:* Since the network is infinite, the resistance of the network to the right of points c and d is also equal to R_T.)

Figure **P26.83**



26.84 ••• Suppose a resistor R lies along each edge of a cube (12 resistors in all) with connections at the corners. Find the equivalent resistance between two diagonally opposite corners of the cube (points a and b in **Fig. P26.84**).

Figure **P26.84**



26.85 ••• BIO **Attenuator Chains and Axons.** The infinite network of resistors shown in Fig. P26.83 is known as an *attenuator chain,* since this chain of resistors causes the potential difference between the upper and lower wires to decrease, or attenuate, along the length of the chain. (a) Show that if the potential difference between the points a and b in Fig. 26.83 is V_{ab}, then the potential difference between points c and d is $V_{cd} = V_{ab}/(1 + \beta)$, where $\beta = 2R_1(R_T + R_2)/R_T R_2$ and R_T, the total resistance of the network, is given in Challenge Problem 26.83. (See the hint given in that problem.) (b) If the potential difference between terminals a and b at the left end of the infinite network is V_0, show that the potential difference between the upper and lower wires n segments from the left end is $V_n = V_0/(1 + \beta)^n$. If $R_1 = R_2$, how many segments are needed to decrease the potential difference V_n to less than 1.0% of V_0? (c) An infinite attenuator chain provides a model of the propagation of a voltage pulse along a nerve fiber, or axon. Each segment of the network in Fig. P26.83 represents a short segment of the axon of length Δx. The resistors R_1 represent the resistance of the fluid inside and outside the membrane wall of the axon. The resistance of the membrane to current flowing through the wall is represented by R_2. For an axon segment of length $\Delta x = 1.0\ \mu\text{m}$, $R_1 = 6.4 \times 10^3\ \Omega$ and $R_2 = 8.0 \times 10^8\ \Omega$ (the membrane wall is a good insulator). Calculate the total resistance R_T and β for an infinitely long axon. (This is a good approximation, since the length of an axon is much greater than its width; the largest axons in the human nervous system are longer than 1 m but only about 10^{-7} m in radius.) (d) By what fraction does the potential difference between the inside and outside of the axon decrease over a distance of 2.0 mm? (e) The attenuation of the potential difference calculated in part (d) shows that the axon cannot simply be a passive, current-carrying electrical cable; the potential difference must periodically be reinforced along the axon's length. This reinforcement mechanism is slow, so a signal propagates along the axon at only about 30 m/s. In situations where faster response is required, axons are covered with a segmented sheath of fatty myelin. The segments are about 2 mm long, separated by gaps called the *nodes of Ranvier*. The myelin increases the resistance of a 1.0-μm-long segment of the membrane to $R_2 = 3.3 \times 10^{12}\ \Omega$. For such a myelinated axon, by what fraction does the potential difference between the inside and outside of the axon decrease over the distance from one node of Ranvier to the next? This smaller attenuation means the propagation speed is increased.

MCAT-STYLE PASSAGE PROBLEMS

BIO **Nerve Cells and *R-C* Circuits**. The portion of a nerve cell that conducts signals is called an *axon*. Many of the electrical properties of axons are governed by ion channels, which are protein molecules that span the axon's cell membrane. When open, each ion channel has a pore that is filled with fluid of low resistivity and connects the interior of the cell electrically to the medium outside the cell. In contrast, the lipid-rich cell membrane in which ion channels reside has very high resistivity.

26.86 Assume that a typical open ion channel spanning an axon's membrane has a resistance of $1 \times 10^{11}\ \Omega$. We can model this ion channel, with its pore, as a 12-nm-long cylinder of radius 0.3 nm. What is the resistivity of the fluid in the pore? (a) $10\ \Omega \cdot \text{m}$; (b) $6\ \Omega \cdot \text{m}$; (c) $2\ \Omega \cdot \text{m}$; (d) $1\ \Omega \cdot \text{m}$.

26.87 In a simple model of an axon conducting a nerve signal, ions move across the cell membrane through open ion channels, which act as purely resistive elements. If a typical current density (current per unit cross-sectional area) in the cell membrane is 5 mA/cm^2 when the voltage across the membrane (the *action potential*) is 50 mV, what is the number density of open ion channels in the membrane? (a) 1/cm^2; (b) 10/cm^2; (c) 10/mm^2; (d) 100/μm^2.

26.88 Cell membranes across a wide variety of organisms have a capacitance per unit area of 1 μF/cm^2. For the electrical signal in a nerve to propagate down the axon, the charge on the membrane "capacitor" must change. What time constant is required when the ion channels are open? (a) 1 μs; (b) 10 μs; (c) 100 μs; (d) 1 ms.

ANSWERS

Chapter Opening Question ?

(ii) The potential difference V is the same across resistors connected in parallel. However, there is a different current I through each resistor if the resistances R are different: $I = V/R$.

Key Example VARIATION Problems

VP26.2.1 (a) 7.00 Ω (b) 0.571 Ω (c) 2.33 Ω (d) 0.857 Ω
VP26.2.2 (a) 8.18 A (b) 6.00 A (c) 2.18 A (d) 2.18 A
VP26.2.3 (a) 54.6 W (b) 20.6 W (c) 18.0 W (d) 16.0 W
VP26.2.4 (a) 9.84 W (b) 5.98 W (c) 2.08 W (d) 1.78 W
VP26.7.1 (a) 1 A (b) 7 V (c) 12 W for 12 V battery, 4 W for 4 V battery
VP26.7.2 (a) 7.75 V (b) 1.25 Ω (c) 0.926 Ω
VP26.7.3 (a) 33.4 W (b) 18.8 W (c) 2.09 W (d) 52.2 W (e) 25.0 W
VP26.7.4 (a) $(2\mathcal{E}_1 + \mathcal{E}_2)/3R$ (b) $-(\mathcal{E}_1 + 2\mathcal{E}_2)/3R$ (c) $(\mathcal{E}_1 - \mathcal{E}_2)/3R$
VP26.13.1 (a) 2.88×10^{-5} C (b) 9.00×10^{-7}A (c) 32.0 s (d) 0.430 (e) 0.570
VP26.13.2 (a) 11.0 s (b) 1.36×10^{-7} A
VP26.13.3 (a) 1.32 MΩ (b) 5.21×10^{-7} A (c) 1.04×10^{-7} A
VP26.13.4 (a) 6.12×10^{-5} C (b) 9.89×10^{-6} C

Bridging Problem

(a) 9.39 J (b) 2.02×10^4 W (c) 4.65×10^{-4} s (d) 7.43×10^3 W